Bird Species Observation Analysis – Project Report

1. Executive Summary

This project analyzes large-scale bird monitoring data to uncover patterns in bird observations, environmental influences, habitat distribution, and conservation priorities.

Using Python for data cleaning and exploratory analysis, SQL for advanced querying, and Power BI for interactive dashboards, the project transforms raw ecological data into meaningful insights that support biodiversity monitoring and conservation decision-making.

2. Problem Statement

Wildlife monitoring programs collect massive observational datasets, but extracting actionable insights from these datasets can be challenging.

Conservation teams and park authorities need a clear understanding of bird species distribution, environmental impacts on bird activity, and identification of watchlist species requiring protection.

Without proper analysis and visualization, it becomes difficult to:

• Identify biodiversity hotspots

• Understand environmental impacts on bird observations

• Detect rare or conservation-priority species

• Support data-driven conservation strategies

3. Business Objectives

The main objectives of this analysis are:

• Analyze bird observation data to identify species distribution patterns.

• Understand how environmental conditions such as temperature, humidity, sky conditions, and wind affect bird observations.

• Identify conservation watchlist species and their observation frequency.

• Detect rare or less frequently observed species that may require conservation focus.

• Provide interactive dashboards that allow stakeholders to explore bird observation trends across parks and habitats.

4. Data Sources

The dataset used in this project consists of two monitoring datasets:

1. Bird Monitoring Data – Forest Habitat

2. Bird Monitoring Data – Grassland Habitat

These datasets contain bird observation records across multiple parks and include environmental conditions recorded during observation.

Key fields include:

• Scientific Name

• Common Name

• Habitat Type

• Admin Unit Code (Park)

• Temperature

• Humidity

• Sky Condition

• Wind Condition

• Environmental Disturbance

• PIF Watchlist Status

5. Data Processing & Cleaning (Python)

Python was used to clean and prepare the dataset for analysis.

Key steps performed:

• Combined multiple sheets from the dataset into a single dataset

• Removed duplicate records

• Handled missing values

• Standardized column formats

• Converted environmental variables into numeric formats

• Created additional analytical fields for visualization

Python libraries used:

• pandas – Data manipulation

• numpy – Numerical operations

• matplotlib – Data visualization

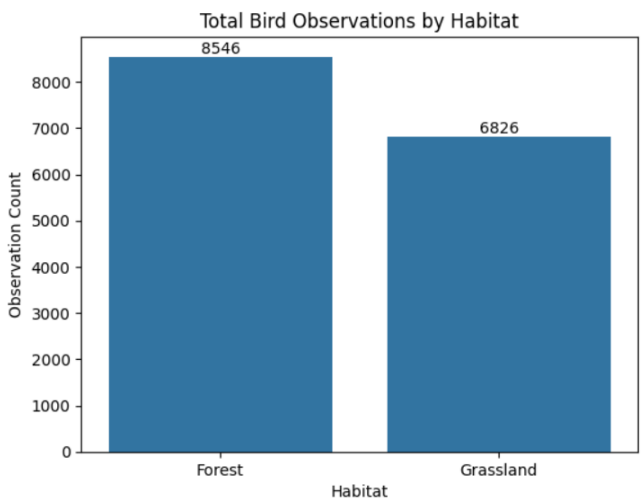
• seaborn – Statistical visualizations

6. Exploratory Data Analysis (EDA)

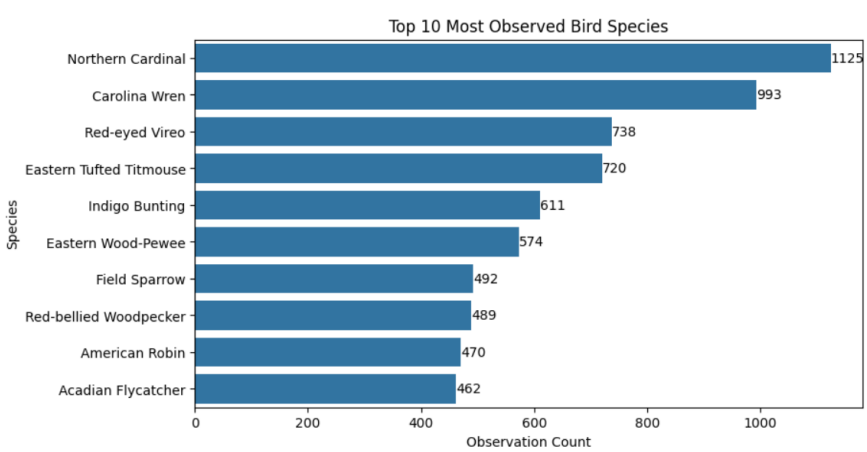
Exploratory Data Analysis was conducted to understand patterns in bird observations.

Key analyses included:

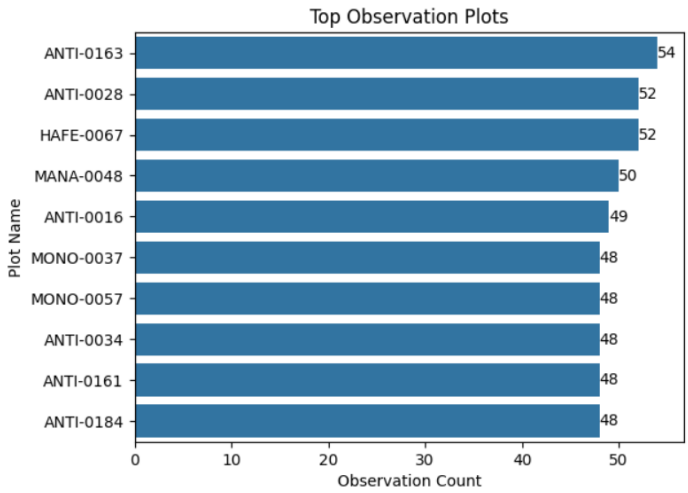
• Bird observations by habitat type



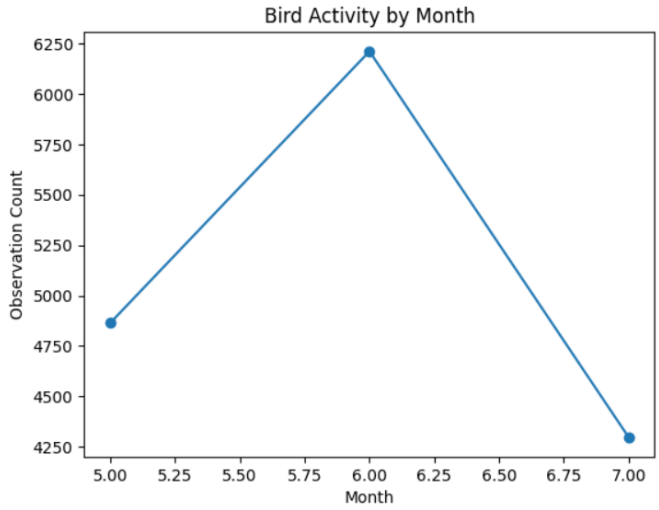
• Top observed bird species



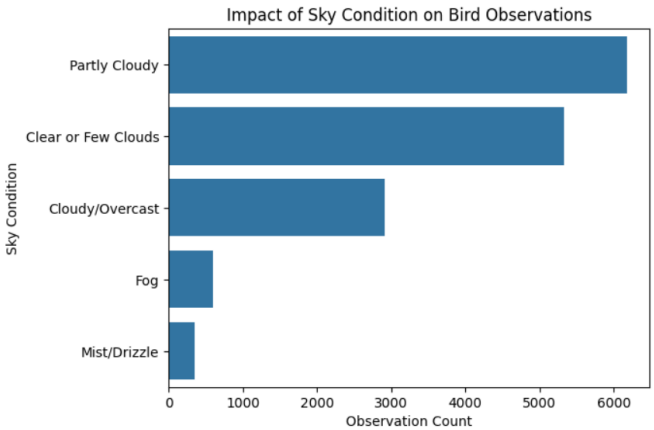
• Top Observation Plots



• Bird observation trends across months



• Impact of Sky Condition on Bird Observations



Visualizations created during EDA helped identify patterns before building dashboards.

7. SQL Data Analysis

After cleaning the data, the dataset was imported into a SQL database using SQLAlchemy.

SQL analysis included:

• Species observation counts

total_observations
15372

• Habitat-wise bird observation analysis

Location_Type	observation_count
Forest	8546
Grassland	6826

• Park-wise biodiversity analysis

Admin_Unit_Code	species_count
MONO	100
ANTI	81
MANA	81
CHOH	80
NACE	66
HAFE	55
PRWI	54
GWMP	49
CATO	46
ROCR	45
WOTR	27

• Watchlist species identification

PIF_Watchlist_Status	species_count
0	14994
1	378

• Average Temperature for Bird Sightings

Location_Type	avg_temperature
Forest	21.87150715319747
Grassland	23.272956332327716

Advanced SQL techniques used:

• Window functions

• Common Table Expressions (CTEs)

• Subqueries

• Aggregation queries

These queries enabled deeper insights into bird distribution and conservation metrics.

8. Power BI Dashboard Development

Power BI was used to build interactive dashboards that present insights in a clear and visual manner.

Three dashboards were created:

Dashboard 1 – Bird Observation Overview

• Total bird observations

• Unique species count

• Observations by habitat

• Top observed bird species

Dashboard 2 – Environmental Impact Analysis

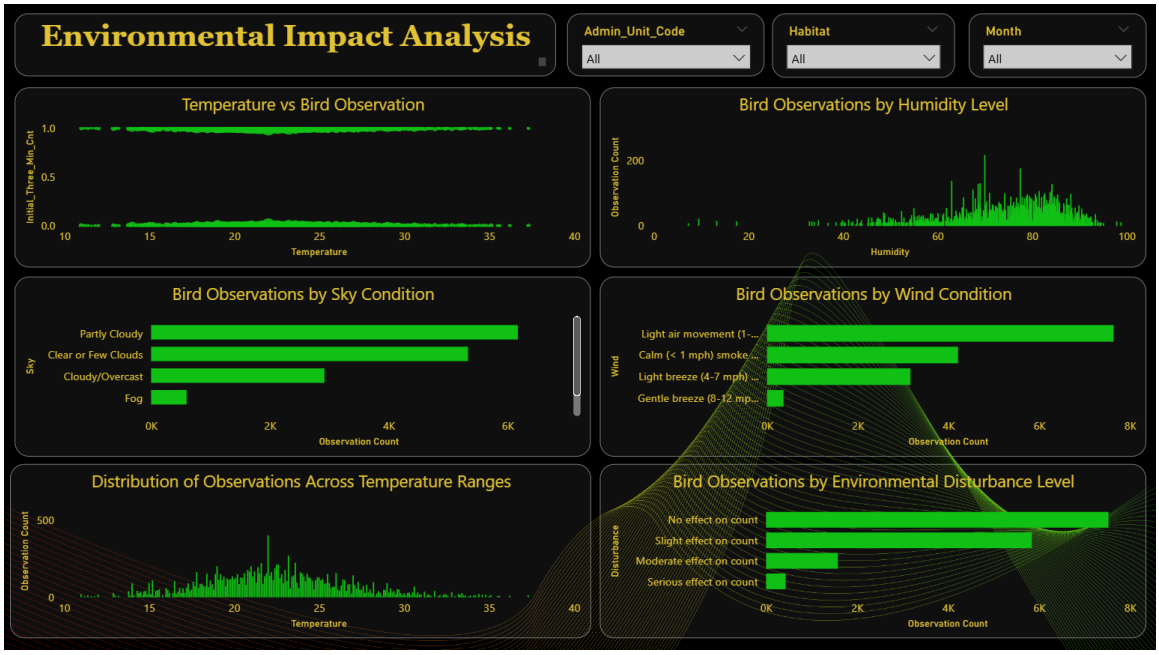
• Temperature vs bird observations

• Humidity distribution

• Bird observations by sky condition

• Bird observations by wind condition

• Environmental disturbance impact



Dashboard 3 – Bird Conservation & Species Insights

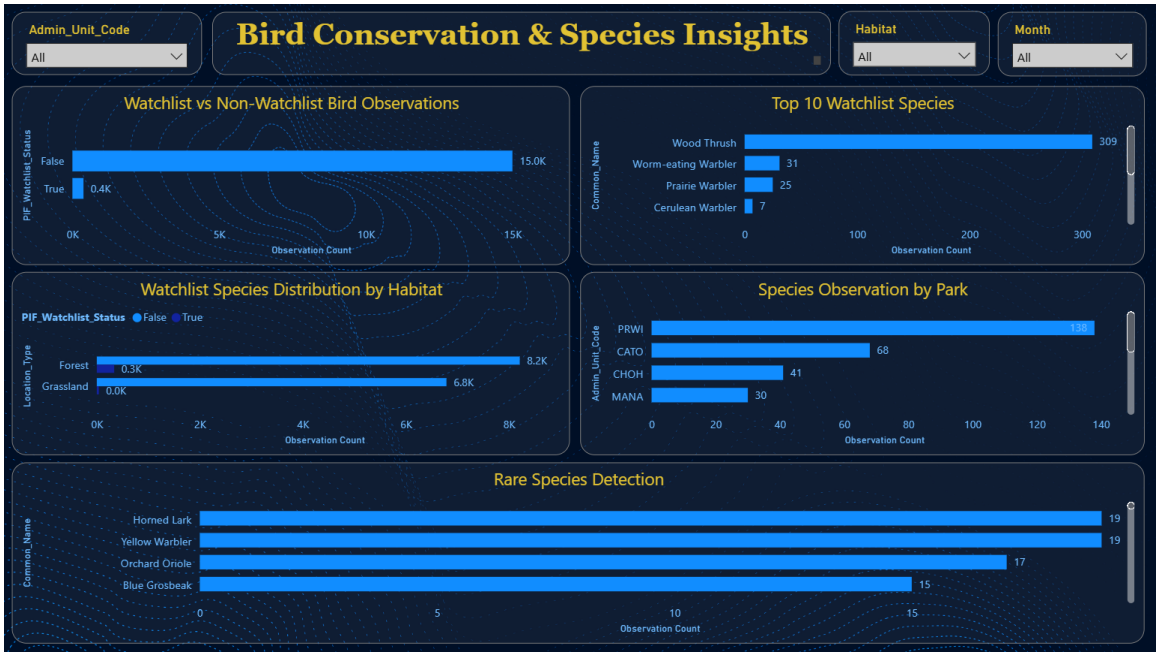
• Watchlist vs non-watchlist observations

• Top watchlist species

• Habitat distribution of watchlist species

• Species observation by park

• Rare species detection



9. Key Insights

Major insights from the analysis include:

• Forest habitats recorded the highest number of bird observations compared to grassland habitats.

• Certain species such as Wood Thrush were frequently observed among watchlist species.

• Bird activity was most common under partly cloudy and clear sky conditions.

• Environmental disturbance had minimal effect on most bird observations.

• Some species were rarely observed, indicating potential conservation priority.

These insights help wildlife managers focus on habitat protection and biodiversity monitoring.

10. Conclusion

This project demonstrates how data analytics can support ecological monitoring and conservation decision-making.

By combining Python, SQL, and Power BI, raw observational data was transformed into meaningful insights that help us to understand bird distribution, environmental influences, and conservation priorities.

The final dashboards provide an interactive platform for exploring bird observation trends and identifying biodiversity hotspots, enabling better wildlife management strategies.